Lecture 18

- Review for midterm 2
- Sections: 14.5, 6
  16.1–4, 5 (2D parts)

Example: $f(x,y,z) = x\cos(\pi y) + ye^z$

$\vec{r}(t)$ curve, $\vec{r}(0) = (2,3,1)$

$\vec{r}\,'(0) = (3,0,4) = \vec{v}$

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$$

$$= \left(\cos \pi y,\ x(-\pi \sin(\pi y)) + e^z,\ ye^z\right)$$

$$\nabla f(2,3,1) = (-1, e, 3e)$$

$$\frac{d}{dt} f(\vec{r}(t))\Big|_{t=0} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}$$

$$= \nabla f(2,3,1)\cdot \vec{r}\,'(0)$$

$$= -1\cdot 3 + e\cdot 0 + 3e\cdot 4$$

$$= -3 + 4e$$

$$\hat{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{(3,0,4)}{5} \quad \text{(unit vector)}$$

$$D_{\hat{u}} f(2,3,1) = \nabla f \cdot \hat{u} = \frac{-3+4e}{5}$$

$D_{\vec{w}} f(2,3,1) = 0$ if

$\vec{w} = (3e, 0, 1)$

Example: $\vec{F} = (ax^2y + y^3 + 1)\hat{i} + (2x^3 + bxy^2 + 2)\hat{j}$

When is $\vec{F}$ conservative?

When $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$, i.e., $\text{curl}(\vec{F}) = 0$.

$$\frac{\partial Q}{\partial x} = 6x^2 + by^2 \qquad \frac{\partial P}{\partial y} = ax^2 + 3y^2$$

$$a = 6,\ b = 3 \iff \vec{F} = \nabla f$$

$$\frac{\partial f}{\partial x} = 6x^2y + y^3 + 1$$

$$\Rightarrow f = 2x^3y + y^3x + x + C(y)$$

$$\frac{\partial f}{\partial y} = 2x^3 + 3y^2x + C'(y)$$

$$\Rightarrow C'(y) = 2 \Rightarrow C(y) = 2y + D$$

C: $x = e^t\cos(t)$ $\quad y = e^t\sin(t)$

$0 \le t \le \pi$

$$\int_C \vec{F}\cdot d\vec{r} = f(\vec{r}(\pi)) - f(\vec{r}(0))$$

$$= f(-e^\pi, 0) - f(1,0)$$

$$= -e^\pi - 1 \;//$$

Force

Example: $\vec{F} = \overbrace{(5x+3y)}^{P}\,\hat{i} + \overbrace{(1+\cos y)}^{Q}\,\hat{j}$

C = unit circle (positively oriented)

Path

$\vec{r}(t) = (\cos t, \sin t) \qquad 0 \leq t \leq 2\pi$

$\langle P, Q\rangle \quad \langle dx, dy\rangle = \hat{T}\,ds$

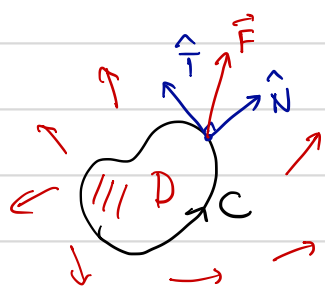
$$\text{Work} = \int_C \vec{F}\cdot d\vec{r}$$

$$= \int_0^{2\pi}\left[(5x+3y)\frac{dx}{dt} + (1+\cos y)\frac{dy}{dt}\right]dt$$

$$= \int_0^{2\pi}\left(5\cos t + 3\sin t\right)(-\sin t) + \left(1+\cos(\sin t)\right)\cos t\,]\,dt$$

$$\text{Work} = \overset{\text{Green's}}{\iint_D} \overbrace{0-3}^{\text{curl}\,\vec{F}}\, dA = -3\pi$$

Green's theorem



circulation of $\vec{F}$ around C

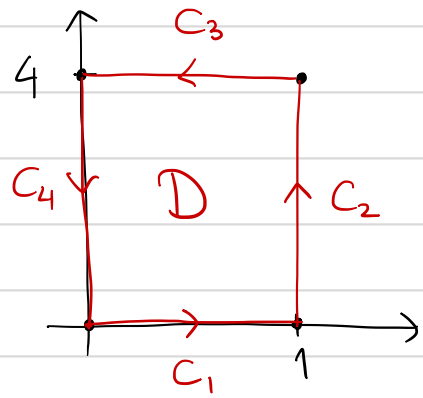
$$\int_C \underbrace{\vec{F}}_{\langle P,Q\rangle}\cdot \underbrace{\hat{T}\,ds}_{\langle dx,dy\rangle = d\vec{r}} = \iint_D \underbrace{\text{curl}(\vec{F})}_{Q_x - P_y}\, dA$$

flux of $\vec{F}$ out of C

$$\int_C \underbrace{\vec{F}}_{\langle P,Q\rangle}\cdot \underbrace{\hat{N}\,ds}_{\langle dy,-dx\rangle} = \iint_D \underbrace{\text{div}(\vec{F})}_{P_x+Q_y}\, dA$$

Example:

$\vec{F} = (xy + \sin x\cos y)\,\hat{i} - (\cos x \sin y)\,\hat{j}$



$C = C_1 \cup C_2 \cup C_3 \cup C_4$

$$\begin{bmatrix}\text{Flux}\\ \text{out of}\\ C\end{bmatrix} = \overset{\text{Green's}}{\iint_R} \overbrace{(y+0)}^{\text{div}\,\vec{F}}\, dA$$

$$= \int_{x=0}^{1}\int_{y=0}^{4} y\,dy\,dx$$

$$= 1\cdot \frac{y^2}{2}\Big|_0^4 = 8$$

(Flux across $C_1 \cup C_2 \cup C_3$)

$$= 8 - \int_{C_4} \vec{F}\cdot\vec{N}\,ds$$

Along C_4: $\vec{F} = \langle 0, -\sin y\rangle$

$\vec{N} = \langle -1, 0\rangle$

$\vec{F}\cdot\vec{N} = 0$

$$\Rightarrow \int_{C_4} \vec{F}\cdot\vec{N}\,ds = 0.$$